

Encoding and modulating

We must transform the data into signals to send them from one place to another. How the information is transformed depends on its original format and on the format used by the communication hardware. **Data Encoding**

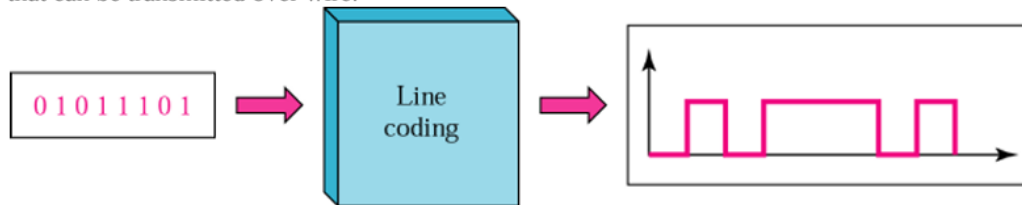
Encoding is the process of converting data from one form to another. Digital data to **digital signal** Digital data to digital signal encoding may be used as the equipment for encoding digital data into a digital signal. Eg: **computer communication using digital line (eg. ISDN)** Digital data to **analog signal** may be used as some transmission media, such as optical fibre and unguided media, will only propagate analog signals Eg: computer communication using voice (telephone) line.

- Analog data to analog signal Analog data and analog signal combination will be used as analog data in electrical form can be transmitted as baseband signals easily and cheaply. Eg: home telephone systems.
- Analog data to digital signal Analog data to digital signal conversion permits the use of modern digital transmission and switching equipment. Eg: digital mobile communication systems.

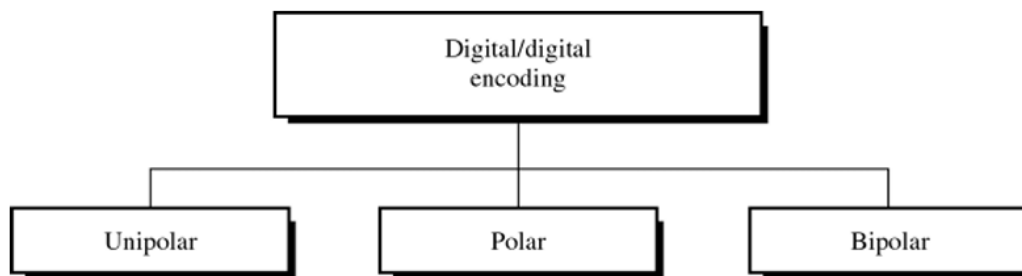
Digital encoding, or line coding, is the process of converting digital data (sequences of 1s and 0s) into a digital signal that can be sent over a physical transmission medium like a cable. The main types of line coding are unipolar, polar, and bipolar, which differ based on the number and polarity of voltage levels they use.

1. Digital to digital encoding

Digital to digital encoding or conversion is the representation of digital information by a digital signal. In this type of encoding the binary 1 and 0 are translated into a sequence of voltage pulses that can be transmitted over wire.



Types of Digital to Digital Encoding



1.1 Unipolar :-

Unipolar Encoding

Unipolar encoding is the simplest method, using only a single voltage polarity to represent data.

- **How it works:** A **positive voltage** represents a binary **1**, and a **zero voltage** represents a binary **0**. The signal never goes into a negative voltage state.
- **Analogy:** Think of an old-fashioned flashlight. When you press the button, the light turns on (a positive state), and when you release it, the light turns off (zero state). The light can't go to a "negative light" state. This is similar to how a unipolar signal works.
- **Real-life example:** Unipolar encoding, particularly in its simplest form (unipolar NRZ), is not widely used in modern, long-distance data communication because of its disadvantages. It's often found in simple, low-power applications like **on-off keying** for short-range infrared communication (e.g., in a TV remote control). The "on" state (light pulse) is a '1', and the "off" state (no light) is a '0'.

Polar Encoding

Polar encoding uses both a positive and a negative voltage to represent data. This method is more efficient than unipolar encoding because it has no DC component (direct current), which is a continuous voltage that can distort the signal and waste power over long distances.

- **How it works:** A **positive voltage** represents a binary **1**, and a **negative voltage** represents a binary **0**.
- **Analogy:** Imagine a seesaw. One side up (positive) represents a '1', and the other side up (negative) represents a '0'. The seesaw is always in motion, transitioning between these two states, but it doesn't just rest flat (at zero) for an extended period like the unipolar system.
- **Real-life example:** A well-known type of polar encoding is **Manchester encoding**, used in early **Ethernet** standards (10BASE-T). In this scheme, a '1' is represented by a transition from a low to a high voltage in the middle of the bit period, while a '0' is a transition from high to low. This design ensures there is always a change in the signal, which helps the receiver stay synchronized with the sender's clock.

Bipolar Encoding

Bipolar encoding uses three voltage levels: positive, negative, and zero. The name "bipolar" is a bit misleading, as it uses three levels, but it gets its name from using two non-zero (positive and negative) voltages to represent a single binary value.

- **How it works:** A **zero voltage** represents a binary **0**. A binary **1** is represented by alternating between a **positive** and a **negative** voltage for each successive '1' bit.
- **Analogy:** Picture a basketball team. The binary '0's are a player sitting on the bench (zero voltage). The binary '1's are two players on the court, alternating who gets the ball (positive voltage, then negative voltage). The action for the '1's always alternates between the two players.
- **Real-life example:** The most common form is **Alternate Mark Inversion (AMI)**, which was used in older telephone networks (like T1 lines). It was

effective at reducing the DC component and was a major improvement over unipolar signaling. However, a long string of 0s can cause a lack of transitions, leading to synchronization problems, which is why more advanced bipolar schemes were developed.

In digital encoding, the three main types—unipolar, polar, and bipolar—describe how a computer's binary data (1s and 0s) are converted into electrical signals. The difference is simply in how many voltage levels they use and how they assign them.

Unipolar Encoding

Unipolar is the most basic method, using only a single voltage level.

- **How it works:** A **positive voltage** is a binary **1**, and a **zero voltage** is a binary **0**. The signal never goes negative.
- **Simple Analogy:** Think of a simple light switch. When you flip it on, it's a "1." When you flip it off, it's a "0." There is no "negative" state for the light.
- **Real-Life Example:** This method is not common for high-speed data because it can cause signal distortion over long distances. However, it's used in very basic applications like on-off keying for a wireless TV remote control, where a pulse of light is a "1" and no light is a "0."

Polar Encoding

Polar encoding uses both positive and negative voltage levels to represent data.

- **How it works:** A **positive voltage** is a binary **1**, and a **negative voltage** is a binary **0**. The signal always changes between these two states, never just staying at zero.
- **Simple Analogy:** Imagine a magnet. You can use the north pole (positive) to represent a "1" and the south pole (negative) to represent a "0." The signal is always "active" in one of these two opposing states.
- **Real-Life Example:** A common type of polar encoding called **Manchester encoding** is used in **early Ethernet** (computer networks). The constant change between positive and negative voltages helps the receiver's clock

stay perfectly in sync with the sender's, which is crucial for high-speed data.

Bipolar Encoding

Bipolar encoding uses three voltage levels: positive, negative, and zero.

- **How it works:** A binary **0** is always a **zero voltage**. A binary **1** is represented by alternating between **positive** and **negative** voltages for each successive '1' bit. For example, if the first '1' is positive, the next '1' must be negative, and the next one after that is positive again.
 - **Simple Analogy:** This is like a coin toss. The "0" is the coin lying flat on the table. The "1"s are represented by the coin landing on either heads (positive) or tails (negative), but they must **alternate** each time a '1' appears. So, a series of '1's would look like: Heads, then Tails, then Heads, then Tails.
 - **Real-Life Example:** A type of bipolar encoding called **Alternate Mark Inversion (AMI)** was used in older telephone systems like **T1 lines**. It was a good way to send data over long distances while preventing signal issues that unipolar and some polar methods have.
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Summary Table

Feature	Unipolar	Polar	Bipolar
Voltage Levels	2 (Positive, Zero)	2 (Positive, Negative)	3 (Positive, Negative, Zero)
How it Works	On/off, like a light switch.	Two opposite states, like magnet poles.	Three states, with '1's alternating polarity.

Feature	Unipolar	Polar	Bipolar
Example Use	Simple TV remotes.	Early Ethernet networks.	Older telephone lines.
Key Benefit	Simple, easy to implement.	Efficient and better for synchronization.	Avoids signal distortion over long wires.

The core difference between **unipolar** and **polar** encoding lies in the number of voltage levels they use and their relationship to the zero-volt line.

Unipolar Encoding

Unipolar encoding is the most basic method, using only a single voltage polarity to represent data.

- **How it works:** A positive voltage typically represents a binary '1', and a zero voltage (no signal) represents a binary '0'.¹ The signal never goes into a negative voltage state. It's like an "on/off" switch.
 - **Drawbacks:** This method has several disadvantages for modern communication.² It has a significant **DC component**, meaning there's a continuous, non-zero voltage, which can cause signal distortion and power loss over long distances.³ It also struggles with **synchronization** because a long string of 0s or 1s results in no voltage change, making it difficult for the receiver to stay in sync with the sender's clock.⁴
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Polar Encoding

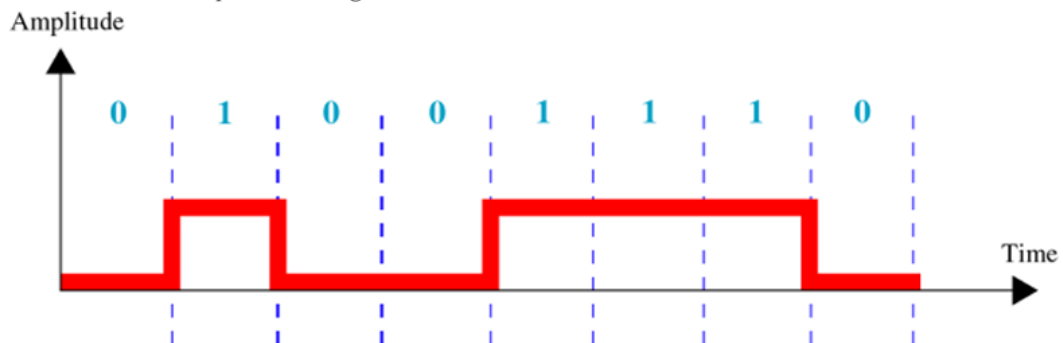
Polar encoding uses both a positive and a negative voltage to represent data.⁵

- **How it works:** A positive voltage represents a binary '1', and a negative voltage represents a binary '0'. The signal always exists in one of these two opposing states. It's like a signal that's always "on," but in two different directions.

- **Advantages:** By using both positive and negative voltages, the signal's average voltage is zero, which means there is **no DC component**.⁶ This makes it much more efficient and reliable for long-distance transmission. Polar encoding also provides better **noise immunity** and can be designed for better synchronization.

Feature	Unipolar	Polar
Voltage Levels	2 (Positive and Zero)	2 (Positive and Negative)
DC Component	High	Zero or very low
Power Efficiency	Less efficient	More efficient
Synchronization	Poor, especially with long strings of identical bits	Better, as transitions often occur more frequently
Noise Immunity	Poor	Good

It uses only one level of value and polarity . Digital transmission system works by sending voltage pulses along a medium link usually cable or wire. The polarity of a pulse refers to whether it is positive or negative.



In this example the 1's are encoded as a positive value and the 0's are encoded as the negative value.

Advantages :-

- Very simple.

Disadvantages :-

- DC component.
- Synchronization.

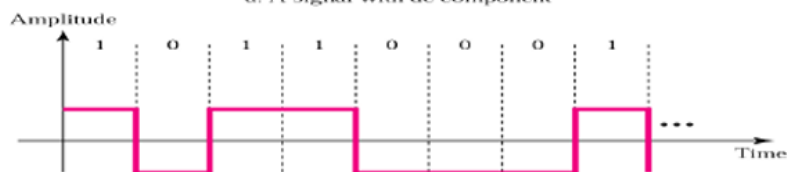
DC component :- When the voltage level in a digital signal is constant for a while, the spectrum creates very low frequencies, called DC components, that present problems for a system that cannot pass low frequencies." Dc is extra energy residing on the line and is useless.

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Note



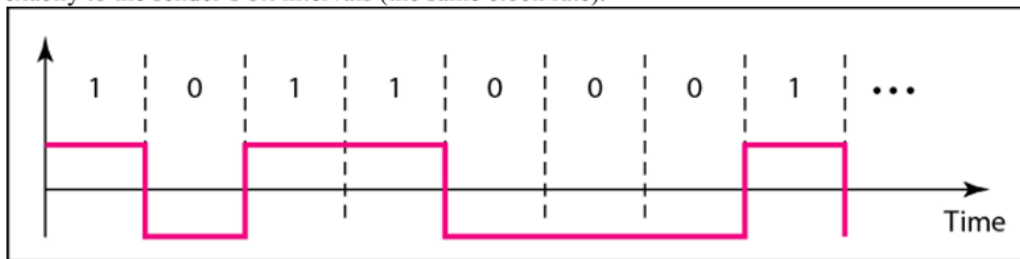
a. A signal with dc component



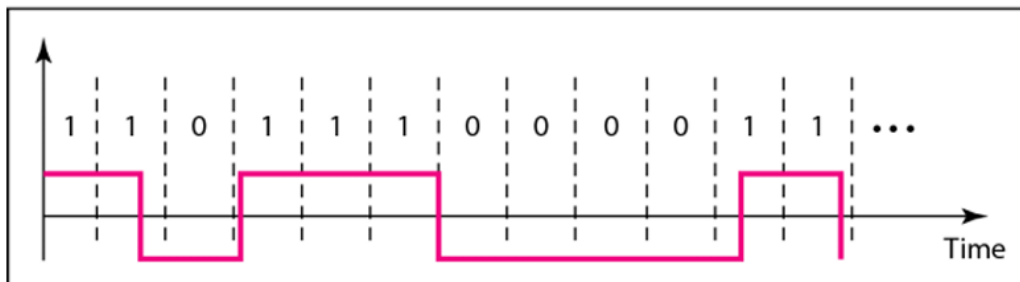
b. A signal without dc component

Synchronization :- when a signal is unvarying, the receiver cannot determine the beginning and ending of each bit. Therefore, a synchronization problem in unipolar can occur when ever the data stream includes a long uninterrupted series of 1's or 0's. Thereceiver receives a continuous voltage and receiver determines how many bits are sent by relaying on its clock (bitrelaying bit--duration), which may not be synchronized with the sender clock.

To correctly interpret the signals received , the receiver's bit intervals must correspond exactly to the sender's bit intervals (the same clock rate).



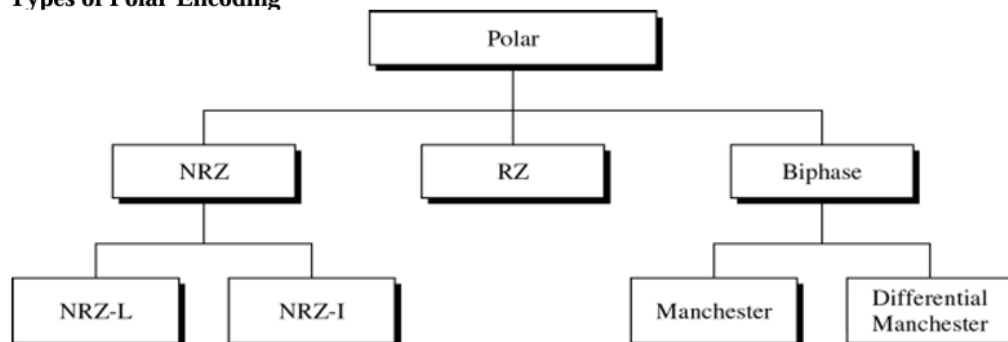
a. Sent



b. Received

1.2 Polar :- It uses two levels of amplitude. One is positive and another is negative, so the DC component problem of unipolar encoding is eliminated.

Types of Polar Encoding



In this encoding technique, the level of the signal is always either positive and negative. It has two types: NRZ-Level, NRZ-Invert.

Non return to zero(NRZ)

NRZ –L :- In NRZ-L the level of the signal is dependent upon the state of the bit. A positive voltage usually means the bit is a zero, and a negative voltage means the bit is one or (vice-versa).

Advantages-Eliminate DC component.

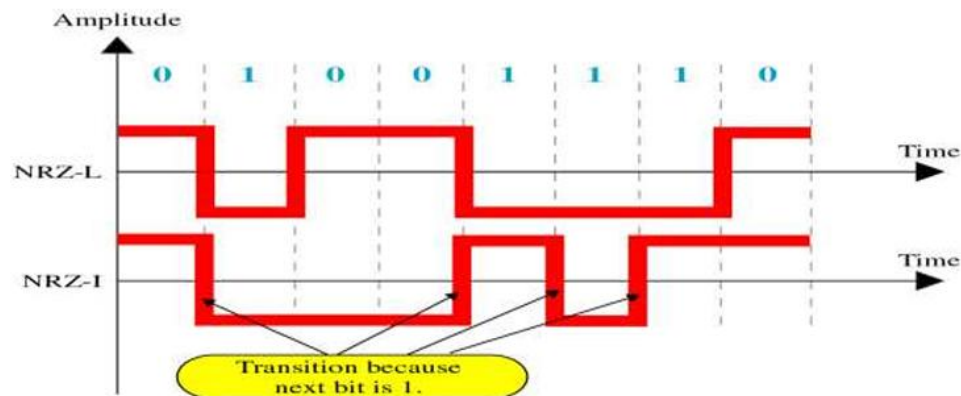
Disadvantages-Lack of synchronization.

Non return to zero(NRZ)

NRZ-I :- In this the signal is inverted if a one is encountered and zero bit is represented by no change.

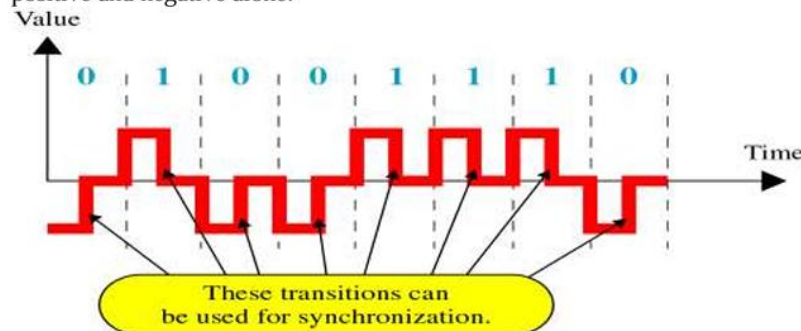
Advantages- Synchronization provided by the signal change.

Disadvantages- String of zeros can still cause problems.



Return to zero (RZ) :- RZ uses three values: positive, negative and zero. In this, the signal changes not between bits but during each bit. A positive voltage means one and negative voltage means zero and halfway through each bit interval, the signal returns to zero.

one bit is represented by positive-to-zero and a **zero bit** by negative-to-zero, rather than by positive and negative alone.



Advantages :-

- Better synchronization.

Disadvantage :-

- It requires two signal changes to encode one bit and therefore occupies more bandwidth.

Biphase

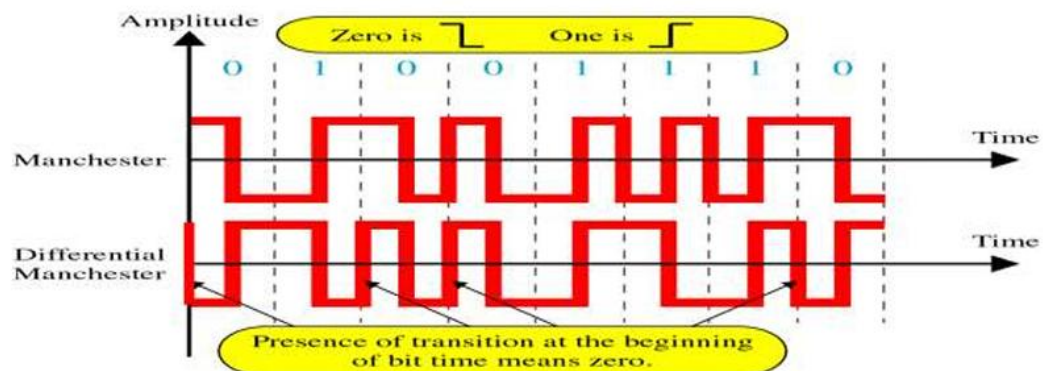
In this method, the signal changes at the middle of the bit interval but does not return to zero.

It can be implemented in two ways:-

- **Manchester**
- **Differential Manchester.**

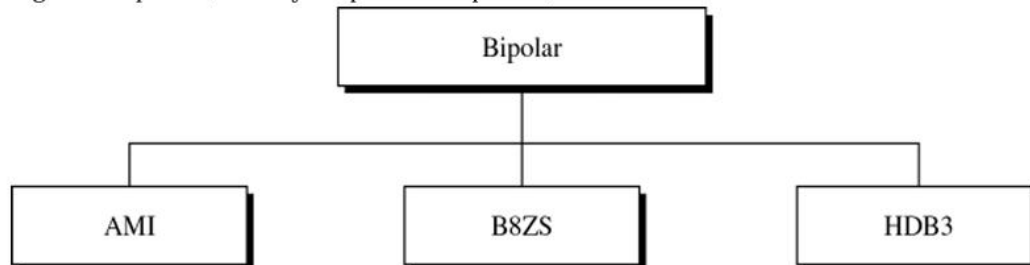
Manchester encoding:- It uses the inversion at the middle of each bit interval for both synchronization and bit representation. A **negative to positive translation** represent **binary one** and a **positive to negative translation** represent **binary zero**. By using the single translation for a dual purpose, manchester encoding achieves the same level of synchronization as RZ but with only two levels of amplitude.

Differential Manchester:- In this, the transition at the middle of the bit is used only for synchronization. The bit representation is shown by the inversion or non inversion at the beginning of the bit. A transition means binary 0 and no transition means binary 1. It requires two signal changes to represent binary 0 but only one to represent binary 1.



1.3 Bipolar encoding :-

It uses three levels of voltage or amplitude: positive, negative and zero. The zero level is used to represent binary zero. The 1's are represented by alternative positive and negative voltages. If the first one bit is represented by the positive amplitude, second will be represented by negative amplitude, third by the positive amplitude, and so on.

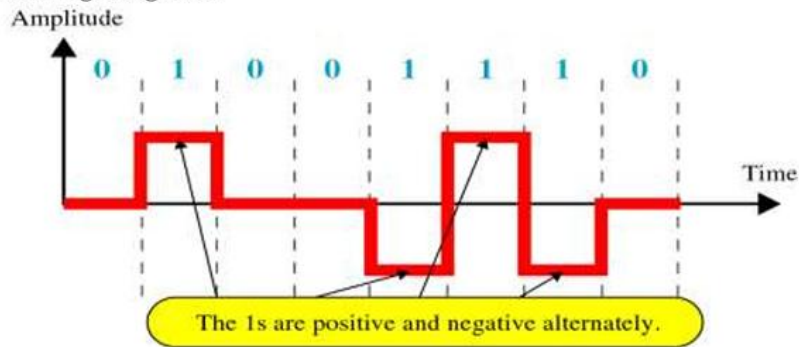


Alternate mark inversion :-

Mark means one, so AMI means alternate one inversion. zero voltage represent binary zero and 1's are represented by alternating positive and negative voltages.

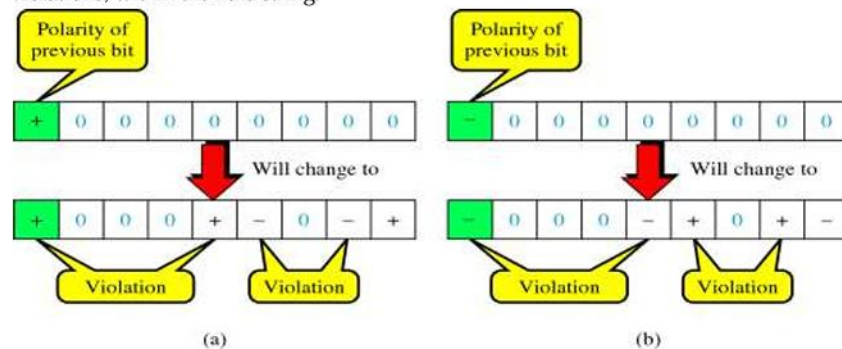
AMI accomplishes two things:

1. DC component is zero.
2. Long sequence of 1s stays synchronization but there is no mechanism to ensure the synchronization of a long string of 0s.



Bipolar 8 zero substitution:-

In this if 8 zeros come one after another, we change the pattern in one of two ways based on the polarity of the previous. The difference between B8ZS and AMI occurs whenever eight or more consecutive 0s are encountered in the data stream. Solution provided by B8ZS is to force artificial signal changes, called violations, within the zero string.



If the previous 1 bit was positive, the eight 0s will be encoded as zero, zero, zero, positive, negative, zero, negative, positive.

When receiver finds two consecutive positive changes surrounding three 0s, it recognizes the pattern as deliberately introduced violation and not an error. It then looks for the 2nd pair of the expected violations. When it finds them, the receiver translates all eight bits to 0s and reverts back to normal bipolar AMI mode. If the polarity of the previous 1 bit is negative, then pattern of violation will be same but with inverted polarities.

High density bipolar-3 :-

In this if four zeros come one after another, we change the pattern based on the polarity of the previous 1 and the number of 1s since the last substitution.

When the no. of 1s since the last substitution is odd, HDB3 Puts a violation in the places of 4th consecutive zeros.

if the polarity of the previous bit was positive, then the violation is positive.

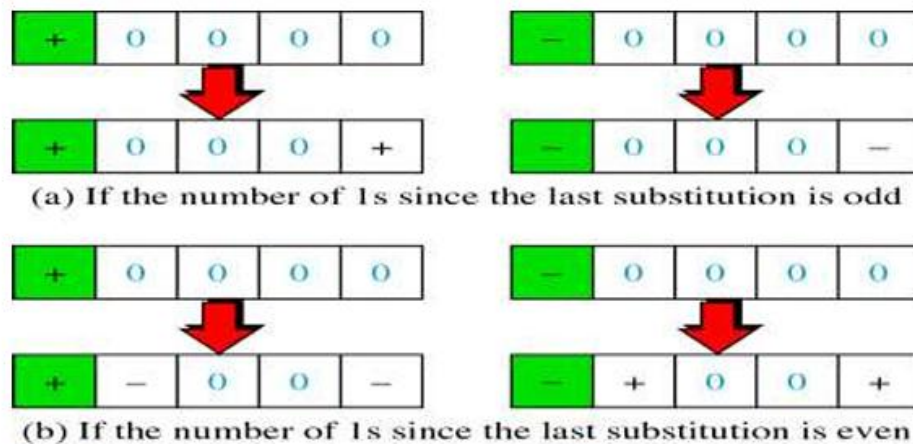
if the polarity of the previous bit was negative, then the violation is negative.

When the no. of 1s since the last substitution is even, HDB3 Puts violations in the places of both the first and the fourth consecutive zeros.

if the polarity of the previous bit was positive, both violations are negative.

if the polarity of the previous bit was negative, both violations are positive.

This was introduced by Europe and Japan and earlier was in US.



Unipolar Encoding

Unipolar encoding is the simplest scheme, using a single voltage polarity.

- **Mathematical Representation:** The signal $s(t)$ for a binary sequence b_i is a series of pulses.
 - $s(t) = A$ for bit '1'
 - $s(t) = 0$ for bit '0'
 - where A is a positive voltage and t is time.
- **Example:** For the binary sequence **10110**, the signal would be:
 - 1: +A
 - 0: 0

- 1: +A
- 1: +A
- 0: 0

Polar Encoding

Polar encoding uses both positive and negative voltage levels. The most common type is **Polar NRZ-L (Non-Return to Zero-Level)**.

- **Mathematical Representation:** The signal $s(t)$ is defined by two voltage levels.
 - $s(t) = +A$ for bit '0' (or '1', depending on convention)
 - $s(t) = -A$ for bit '1' (or '0')
 - The voltage level remains constant for the entire bit duration.
- **Example:** For the binary sequence **10110**, assuming '1' maps to +A and '0' maps to -A, the signal would be:
 - 1: +A
 - 0: -A
 - 1: +A
 - 1: +A
 - 0: -A

Bipolar Encoding (Alternate Mark Inversion - AMI)

Bipolar AMI uses three voltage levels: positive, negative, and zero. The '1' bits alternate between positive and negative voltages, while '0' is always zero.

- **Mathematical Representation:** The signal $s(t)$ has three states.
 - $s(t) = 0$ for bit '0'
 - $s(t) = +A$ or $-A$ for bit '1', with the polarity alternating with each successive '1'.
- **Example:** For the binary sequence **101101**, the signal would be:
 - 1: +A
 - 0: 0

- 1: -A (the polarity alternates from +A)
- 1: +A (the polarity alternates from -A)
- 0: 0
- 1: -A (the polarity alternates from +A)

Manchester Encoding

Manchester encoding combines clocking and data into a single stream. A transition in the middle of each bit period is used for synchronization.

- **Mathematical Representation:** The signal $s(t)$ is a function of the data bit b_i and the clock signal.
 - For bit '0': $s(t)$ transitions from **high to low** in the middle of the bit period.
 - For bit '1': $s(t)$ transitions from **low to high** in the middle of the bit period.
- **Example:** For the binary sequence **10110**, the signal transitions would be:
 - 1: Low-to-high transition
 - 0: High-to-low transition
 - 1: Low-to-high transition
 - 1: Low-to-high transition
 - 0: High-to-low transition

Alternate Mark Inversion (AMI)

This is a specific type of Bipolar encoding, so its mathematical representation and example are the same as described above. A key mathematical characteristic of AMI is that the average DC (direct current) component is close to zero, which is a significant advantage for long-distance transmission over a transformer-coupled line. The alternating polarity of the '1's ensures that the positive and negative voltage pulses cancel each other out over time.

Comparison of Encoding Schemes

Feature	Unipolar NRZ	Polar NRZ-L	Bipolar AMI	Manchester
Voltage Levels	2 (Positive & Zero)	2 (Positive & Negative)	3 (Positive, Negative, Zero)	2 (Positive & Negative)
DC Component	High	Zero	Zero	Zero
Bandwidth	Lowest	Lowest	Low (less than NRZ)	Highest (Twice that of NRZ)
Synchronization	Poor (long runs of 1s/0s)	Poor (long runs of 1s/0s)	Poor (long runs of 0s)	Excellent (self-clocking)
Error Detection	None	None	Yes (for polarity violation)	Yes (for missing transition)

- Bit Interval and Pulse Width:** A signal uses **Unipolar RZ** (Return to Zero) encoding at a bit rate of 2 Mbps.
 - **Question:** What is the duration of a single bit interval in microseconds? What is the pulse width for a binary '1' in this encoding scheme?
- Average Power (Polar NRZ):** A **Polar NRZ-L** signal with a voltage of +3V for a binary '1' and -3V for a binary '0' is used to transmit the bit stream **10101010**. Assuming the resistance of the channel is 50 ohms, calculate the average power of the signal.

3. **Synchronization and Transitions:** A system transmits the binary sequence **111000101** using a **Differential Manchester** encoding. Assuming the first bit starts with a transition, calculate the total number of transitions (at the beginning of the bit and in the middle) in the signal.
4. **Signal Levels in Pseudoternary:** A **Pseudoternary** encoding scheme uses three voltage levels: positive (+V), negative (-V), and zero (0V). How is a binary '1' represented in this scheme? How is a binary '0' represented? Explain how this differs from **Bipolar AMI**.
5. **Bandwidth Efficiency:** A **Polar NRZ-L** scheme has a maximum bit rate of 100 Mbps on a channel with a specific bandwidth. If the same channel is used for **Manchester encoding**, what is the maximum achievable bit rate? Explain why there is a difference.
6. **Clocking in Biphase Encoding:** Explain the primary reason why **Manchester** and **Differential Manchester** encoding schemes are considered "self-clocking." Use a mathematical or technical explanation related to signal transitions.
7. **Noise Immunity and Signal-to-Noise Ratio (SNR):** A **Polar NRZ-L** signal uses voltage levels of +5V and -5V. A noise signal with an amplitude of 1V is introduced.
 - **Question:** What is the signal-to-noise ratio (SNR) in decibels (dB)? How does this relate to the noise immunity of the encoding scheme?
8. **Error Detection in Bipolar AMI:** The binary sequence **1011001** is transmitted using **Bipolar AMI** encoding. The first '1' is positive.
 - **Question:** What is the expected voltage pattern for this sequence? If the receiver detects a pattern where a positive pulse is immediately followed by another positive pulse, what does this indicate?
9. **Hamming Distance and Error Detection (Revisited):** In a simplified encoding, two codewords 010 and 111 are used.
 - **Question:** Calculate the Hamming distance between these two codewords. Based on this distance, what is the maximum number of bit errors that can be detected? What is the maximum number of errors that can be corrected?

10. Calculating Bit Rate from Baud Rate: A modulation scheme transmits 4 bits per signal change (symbol). If the Baud rate is 2400 Baud, what is the bit rate in bps? Explain the relationship between bit rate and baud rate in this context.